

GuessScape: A Comparative Study of Playful and Gameful Elements in Smart Mirror-Based Self-Reflection

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Abstract

Smart mirrors integrate self-reflection into daily grooming routines, yet keeping that practice both engaging and meaningful remains a challenge. This paper tackles this challenge by presenting GuessScape, which is a smart mirror system that replays wearable camera photos taken throughout the day and adds two optional elements: (i) a playful quiz based on the captured photos and (ii) a gameful leaderboard. The quiz fosters curiosity-driven guessing without scoring, whereas the leaderboard introduces household competition. GuessScape offers three modes, Replay, Playful, and Gameful, to compare the effects of playful and gameful elements. This is evaluated through a user-enactment study combining the Game Experience Questionnaire with reflection items. Results indicate that both Playful and Gameful modes foster deeper reflection, improve well-being, and provide a more positive user experience than Replay. However, while some participants appreciated the competition in the Gameful mode to increase their motivation, several also noted that it risked turning the reflective process into a mere score-chase, highlighting the advantage of playful elements in sustaining meaningful self-reflection.

CCS Concepts

• **Human-centered computing** → **Ubiquitous computing**; **Gamification**; *User studies*; *Empirical studies in HCI*.

Keywords

Ubiquitous Computing, Smart Mirror, Gamification, Well-being, User Engagement, Everyday Reflection, Interaction Design

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1 Introduction

Ubiquitous computing artifacts that encourage self-reflection have been associated with greater psychological well-being and sustainable behavior change [12]. In this context, self-reflection refers to an intrinsically motivated, open-ended, and personally oriented practice that revisits one's own everyday life [8]. Recent work has explored how smart mirrors, mirrors augmented with displays and sensors, can serve as a self-reflection device [3, 9, 19, 23]. This is due to their ability to embed reflective prompts into pre-existing everyday grooming routines, thereby lowering the effort barrier that often hampers mobile or desktop-based solutions [4, 14].

While lowering the effort barrier is necessary, it does not automatically make the reflection process appealing. The quality of self-reflection experience is critical; in one study on the use of a self-reflection system, 23.8% of participants dropped out within six months, despite user engagement being a key factor for the success of such systems [1]. Accordingly, our focus shifts to how to improve the quality of the reflective experience.

A common method to enhance user experiences is gamification, which is the use of game elements, such as scores and leaderboards, in non-game contexts [6]. In fact, a systematic review concluded that gamification can increase short-term engagement across on-line interventions [15]. However, prior research has cautioned that poorly integrated gamification elements can undermine the intended purpose. One example is a breastfeeding training game in which scores diverted participants' focus from nursing techniques to point acquisition, thereby undermining the intended reflective learning outcomes [21]. Yet these findings do not imply that all game elements are detrimental. Rather, they expose an unresolved question: how gamification elements can be integrated to motivate sustained engagement while supporting meaningful reflection.

To explore this knowledge gap, this paper introduces GuessScape (Guess + Landscape). This is a smart mirror system that encourages self-reflection through revisiting random landscapes that users encounter during the day, which are automatically captured. The system features three modes based on Bentvelzen et al's four design resources for reflection [2]: Replay Mode, Playful Mode, and Gameful Mode. In Replay Mode, the system replays the captured images on the mirror display as a baseline. In Playful Mode, GuessScape hides one randomly chosen object per image using object detection and generates a three-choice quiz. In Gameful Mode, the system introduces the quiz with a household leaderboard to encourage competition. This paper refers to the quiz as a playful element and the leaderboard as a gameful element.

GuessScape is evaluated through a user-enactment study [16] that contrasts the three prototypes to answer two questions. RQ1:

How do playful and gameful elements differently influence self-reflection with a smart mirror? RQ2: Is there a trade-off between player experience and the depth of reflection when these elements are introduced in self-reflection with a smart mirror? In the user study, fifteen participants interacted with GuessScape through a prescribed scenario and completed the Game Experience Questionnaire (GEQ) and self-reflective questions [11]. This setup enables within-subject comparisons of engagement and reflective depth.

This paper makes two contributions. First, it introduces GuessScape, the first mirror-based system that integrates playful and gameful elements to support daily reflection. Second, this paper provides empirical evidence comparing engagement and reflective quality across Replay, Playful, and Gameful modes, providing insights for the design of future ubiquitous-computing artifacts that balance motivation with meaningful reflection.

2 Related Work

2.1 Smart Mirrors for Everyday Reflection

Smart mirrors have been explored as reflection devices that can be embedded into everyday grooming routines with minimal additional effort. For example, Mirror Ritual is a mirror-based system that turns facial expression classifications into generative poetry, prompting users to reframe their emotions [19]. Building on this idea, LUX augments the mirror with context-sensitive responses to comfort users [23]. Also, Bianco et al. integrate multimodal affect detection and emotion-trend charts in a household mirror [3]. Most recently, Baracoda's BMind prototype couples generative-AI conversation, sentiment recognition, and light therapy guidance to support daily mental wellness coaching [9].

While these smart mirror systems excel at capturing users' momentary emotional states, they overlook situated memory cues evoked by re-encountering scenes from one's own day. A previous visual-lifelog study shows that reviewing wearable-camera photos each evening can spark spontaneous reflection and help users make sense of daily activities without substantial effort overhead [13]. GuessScape builds on this insight by replaying same-day images in the mirror and layering interactive prompts that aim to deepen, rather than interrupt, that everyday meaning-making.

2.2 Gamified Engagement & Reflective Depth

Gamification, the addition of game-like elements such as points and leaderboards to non-game activities, is a widely adopted strategy for increasing engagement in ubiquitous computing systems [6]. For instance, a systematic review of 1,017 studies confirms that such elements are generally effective in raising engagement [15].

Yet, stronger engagement does not always lead to deeper reflection. When external rewards dominate, users may shift their focus to gaining points instead of pursuing insights. For example, students using a gamified e-learning platform noted that moving up a leaderboard does not necessarily mean people know more [7]. Likewise, students in a gamified class tended to have less intrinsic motivation [5], resulting in lower exam scores [10]. Similarly, a breastfeeding training game drew participants' focus to scores rather than the infant avatar, diverting attention from proper technique [21]. These cases illustrate that careful integration is essential when introducing gamification into self-reflection contexts.

In terms of design principles for self-reflection systems, Bentvelzen et al. identify four design resources that can encourage reflection: temporal perspective, discovery, comparison, and conversation [2]. GuessScape was designed with these resources in mind, as explained in the following sections.

3 System Design

3.1 Overview

GuessScape is implemented as an interactive smart mirror application, running on a Windows tablet for ease of development and testing. Utilizing the tablet's camera, the system continuously captures and renders live video as the background layer, producing a convincing mirror-like effect. An external wearable camera, worn by the user during daily activities, captures images at randomized timings. The entire software stack is developed in Unity (version 6000.0.47f1), which integrates image processing, interaction control, and API communication [22].

3.2 User Interaction Modes

GuessScape supports three interaction modes, shown in Figure 1.

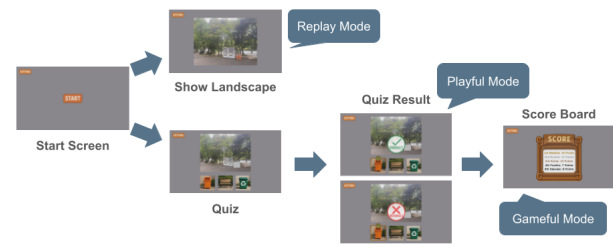


Figure 1: Interaction modes and transitions in GuessScape

- (1) **Replay Mode:** A non-interactive mode in which the captured image is shown as-is. This acts as a passive reflection tool, presenting visual memories without additional stimuli.
- (2) **Playful Mode:** The user is presented with an object-hiding quiz. Upon selecting one of the images, the correct answer is revealed. No scores are shown, promoting open-ended exploration rather than competitive behavior.
- (3) **Gameful Mode:** This is similar to the Playful Mode but allows users to view a family leaderboard based on fictional members within the scenario.

Note that the background surroundings captured by the camera are hidden in the figure for privacy reasons. The design of these modes is based on Bentvelzen et al.'s four design resources for reflection, which are *temporal perspective*, *discovery*, *comparison*, and *conversation* [2]. Replay Mode uses *temporal perspective* by reappearing the day's photographs on the mirror. Playful Mode adds object-hiding quizzes without scores, *discovery* fostered through curiosity-driven guessing. Gameful Mode layers the same quiz with points and a household leaderboard, activating *comparison* and potentially sparking *conversation* among family members.

3.3 Data Acquisition and Processing Pipeline

Figure 2 illustrates the data flow and software architecture. In short, GuessScape operates through the following data-processing loop:

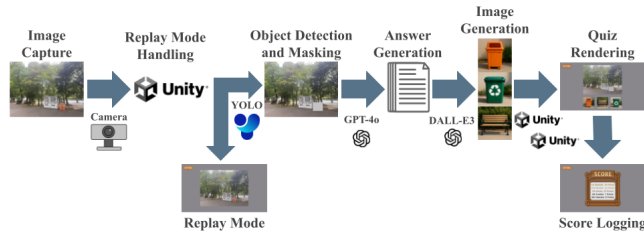


Figure 2: System configuration and data flow of GuessScope

- (1) **Image Capture:** Photos are captured automatically from a wearable camera at random times during the day.
- (2) **Replay Mode Handling:** In Replay Mode, the captured image is displayed at the center of the screen and finishes.
- (3) **Object Detection and Masking:** For Playful and Gameful modes, the image is processed with on-device object detection using YOLO, a real-time object detection model [20]. A list of detected objects and their bounding boxes is extracted. Then, one object is randomly selected and its corresponding bounding box is overlaid with a mask.
- (4) **Answer Generation:** The label of the masked object is sent to GPT-4o API [18], a large language model capable of generating text based on contexts, to retrieve the names of two objects that might be found in a similar environment.
- (5) **Image Generation:** Using DALL-E3 API [17], a text-to-image generative AI model, three candidate images are generated: one depicting the correct object and two depicting the incorrect objects suggested by GPT-4o.
- (6) **Quiz Rendering:** The scene with the masked object and the three candidate object images is displayed together. The user selects an answer and the correct answer is indicated.
- (7) **Score logging:** For Gameful Mode, each user's responses are tracked and scored throughout the session. These scores are stored locally under the active user profile.

To ensure privacy, only object labels are sent to external APIs; the actual captured images are never uploaded. Additionally, a hardware switch on the camera allows participants to manually pause image capture in private situations.

4 Experiment

4.1 Research Questions

The experiment addressed two research questions:

- **RQ1:** How do playful and gameful elements differently influence self-reflection with a smart mirror?
- **RQ2:** Is there a trade-off between player experience and the depth of reflection when these elements are introduced in self-reflection with a smart mirror?

4.2 Participants

Fifteen university students (7 female, 8 male; $M = 22.5$ years, $SD = 1.4$) participated in the experiment. All were students of the university where the datasets were captured and regularly commuted through those places, ensuring familiarity with the depicted scenes.

4.3 Procedure

A user-enactment method [16] was used; each participant enacted a scenario in which a commuter student reviewed photos with GuessScope during evening tooth-brushing for self-reflection. Three fixed datasets were used; the photo sets were captured in advance by the first author wearing a wearable camera during routine walks around the campus, which matches the scenario depiction. Object detection, masking, and candidate-object generation were likewise pre-computed to normalize quiz difficulty across participants. The datasets used in the experiment are shown in Figure 3.

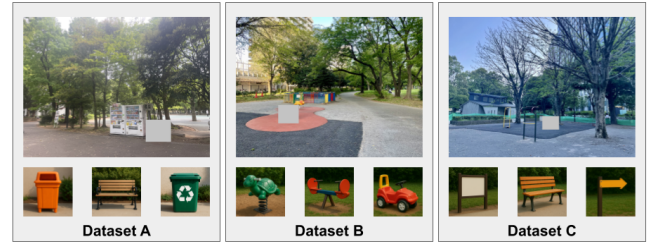


Figure 3: The three datasets used in the experiment

A Latin-square design counterbalanced both mode order and dataset assignment; each participant was randomly assigned to one of the mode sequences, so every mode and dataset appeared in every ordinal position exactly once, neutralizing order effects.

In short, the flow of the experiment was as follows:

- (1) **Briefing & Consent:** Explained the study's goal, obtained informed consent, and introduced the prescribed scenario.
- (2) **System Explanation:** Explained the GuessScope mirror interface, quiz interaction, and leaderboard.
- (3) **Pre-study Questionnaire:** Collected demographics and assessed participants' familiarity with the route.
- (4) **Enactment Blocks → Per-mode Questionnaire:** Participants experienced GuessScope, followed by the per-mode questionnaire after each mode: GEQ, two validated reflection items ("encouraged self-reflection", "improved daily well-being") rated on a 5-point Likert scale, followed by open-ended responses explaining the ratings.
- (5) **Post-study Questionnaire:** Open-ended responses about participants' feedback and preferred mode.

5 Result and Discussion

5.1 RQ1: Influence of Playful and Gameful Elements on Reflection

All participants visited the dataset locations at least once a week, except for one participant, who visited them monthly, showing the relevance of datasets. As summarized in Figure 4, in the Gameful Mode, 12 out of 15 participants rated their reflection experience 4 or higher, with an average score of 4.33. Similarly, in the Playful Mode, 14 participants gave ratings of 4 or higher with an average of 4.13. In contrast, only 4 participants rated the Replay Mode 4 or higher, with a considerably lower average score of 2.53. In Figure 5, 12 participants in Gameful Mode and 11 in Playful Mode rated the enhancement of well-being 4 or higher, with average scores of 4.07

and 3.87, respectively. In contrast, the Replay mode was rated 4 or higher by only 6 participants, with an average score of 2.87.

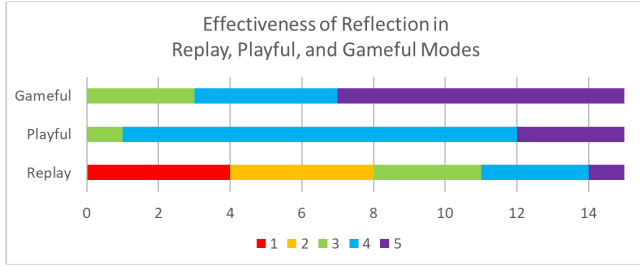


Figure 4: Distribution of the 15 participants' ratings on the item "The system encouraged self-reflection". Each horizontal bar shows how many participants selected each point on the 5-point Likert scale (1 = strongly disagree, 5 = strongly agree); the x-axis therefore ranges from 0 to 15.

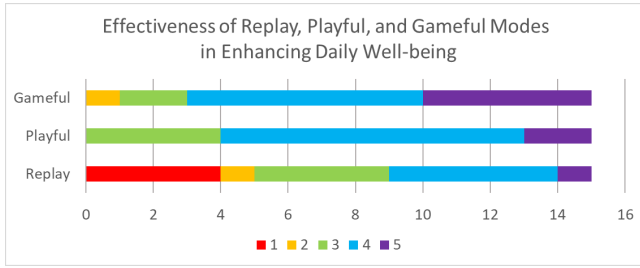


Figure 5: Distribution of the 15 participants' ratings on the item "The system improved my daily well-being". Color coding and x-axis meaning are identical to Figure 4.

Qualitative responses from participants (P1–P15) reinforce the quantitative findings. Participants found the Replay mode “dull” (P2) and “lacking focus” (P9), whereas the Playful mode “drew attention to unnoticed details” (P11) and “offered a reason to observe more closely” (P10). These comments illustrate the effects of playful elements on reflection: by prompting users to guess hidden objects, the system redirected attention toward familiar environments with fresh curiosity. In addition, the result shows that the gameful elements added a social dimension. Participants described being “motivated by the scores” (P4) and “excited to see the results of family members” (P9). However, not all reflections were positive; while some valued increased motivation through competition (P2, P5, P9), others noted that the leaderboard led them to focus excessively on scores, detracting from the reflective experiences (P7, P8, P10). Overall, the results illustrate that incorporating playful and gameful elements makes the mirror experience substantially more reflective and emotionally positive than simply replaying images.

5.2 RQ2: Trade-off Between Player Experience and Reflective Depth

Table 1 summarizes the mean GEQ scores per mode. Statistical tests were conducted on each participant’s raw scores. A Friedman test

revealed significant differences ($p < 0.05$) among modes for all GEQ categories except *Tension* and *Negative Affect*. Bonferroni-corrected Wilcoxon signed-rank comparisons ($\alpha = 0.017$) showed that both Playful and Gameful modes produced higher *Competence*, *Challenge*, and *Positive Affect* than Replay; Gameful also exceeded Replay on *Immersion*. No reliable differences emerged between Playful and Gameful. Thus, adding either playful or gameful elements boosted users’ sense of skill, absorption, and enjoyment without increasing tension or negative affect.

Table 1: Average GEQ Scores Across Modes. Bold values differ significantly from Replay according to Bonferroni-corrected Wilcoxon tests. (A: Competence, B: Sensory and Imaginative Immersion, C: Flow, D: Tension/Annoyance, E: Challenge, F: Negative affect, G: Positive affect).

	A	B	C	D	E	F	G
Replay	2.31	2.04	1.44	1.29	1.29	1.73	2.48
Playful	3.12	2.73	2.91	1.36	1.89	1.35	3.64
Gameful	3.07	2.91	2.01	1.31	1.79	1.44	3.68

The critical contrast between playful and gameful elements, therefore, lies not in the overall experience but in where users direct their attention. Qualitative comments reveal a split: some participants reported that the leaderboard “motivated closer observation” (P5), whereas others felt it “drew focus to the scores, away from the self-reflection” (P8). Ten of fifteen participants selected Gameful as their favorite mode, yet five preferred Playful, suggesting that competition can enhance engagement for some users while impairing self-reflection for others. Designers should therefore weigh these motivational benefits against the risk of score-chasing that the statistical data alone cannot capture.

6 Conclusion and Future Directions

This paper presents GuessScape, a system that advances smart mirror self-reflection by contrasting three modes: Replay, Playful, and Gameful. The user study showed that both playful and gameful elements significantly outperformed Replay Mode on reflection items and well-being, while also enhancing player experience. The leaderboard in Gameful Mode did not significantly affect GEQ scores but, according to several participants, sometimes redirected attention from the scene to the score or increased motivation. These confirm that playful elements reliably deepen reflection, whereas gameful elements can be motivating yet may compromise reflective depth.

Future work should deploy GuessScape longitudinally in domestic bathrooms to test whether the benefits of playful and gameful elements persist once novelty declines. Also, broader household samples are needed to examine how age, family dynamics, and prior gaming experience influence preferences for each mode. Finally, adaptive controls that let users adjust quiz difficulty and toggle the leaderboard visibility could help balance motivation with meaningful self-reflection, informing the design of future ubiquitous systems aimed at embedding reflection into everyday routines.

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